

Experience with Carbon Taxes and Greenhouse Gas Emissions Trading Systems

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Erik Haites, Margaree Consultants Inc., Toronto, Canada*

Duan Maosheng, Tsinghua University, Beijing, China

Kelly Sims Gallagher, Tufts University, Medford, Massachusetts, USA

Sharon Mascher, University of Calgary, Calgary, Alberta, Canada

Easwaran Narassimhan, Tufts University, Medford, Massachusetts, USA

Kenneth R. Richards, Indiana University, Bloomington, Indiana, USA

Masayo Wakabayashi, Central Research Institute of Electric Power Industry, Tokyo, Japan

* Corresponding author: Ehaites@margaree.ca

Abstract

Carbon taxes and emissions trading systems (ETs) to limit emissions of greenhouse gases (GHGs) are becoming increasingly common. At the end of 2015, 17 GHG ETs were operational in 55 jurisdictions while 18 jurisdictions collected a carbon tax. Empirical evidence on the performance of carbon taxes and GHG ETs is sparse. This paper assesses the performance of those taxes and ETs as implemented with respect to: environmental effectiveness (reduction of actual emissions), cost-effectiveness (marginal abatement cost), economic efficiency, public finance and administrative issues. Actual emissions fell for seven of 13 carbon taxes for which data are available and for at least six of ten ETs. Taxes and ETs in other jurisdictions may have reduced emissions from what they would have been but not enough to lower actual emissions. In at least 3 tax jurisdictions, the emission reductions appear to be due to complementary policies. Data to determine the contribution of complementary policies to the reduction of emissions covered by ETs are not available. On average the marginal cost is substantially lower for ETs than carbon taxes. The emissions reductions achieved by existing carbon taxes are small in most jurisdictions due to the low tax rates, the modest changes in tax rates and inelastic demands for fossil fuels. Existing taxes yield virtually no insight into the relationship between changes to the tax rate and changes to emissions. All of the ETs have accumulated banks of surplus allowances and most have implemented measures to reduce these banks.

Both instruments can reduce emissions but ETs perform better than carbon taxes on the principal criteria of environmental effectiveness and cost-effectiveness. More research is needed to improve the design of carbon taxes and GHG ETs to make them more effective and to inform the key decisions policymakers need to make when implementing them; setting the tax rate for a carbon tax and adjusting it over time and setting the emissions cap for an ET and adjusting it over time.

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1. Introduction

The relative merits of a carbon tax and a greenhouse gas emissions trading system (GHG ETS) as policies to reduce greenhouse gas emissions have been widely debated in the literature.¹ Much of the debate is theoretical because empirical evidence relating to the performance of these instruments is sparse. This paper seeks to advance our understanding of carbon taxes and GHG ETSs by focusing on their performance as implemented.

Economic theory indicates that pricing² GHG³ emissions is the most economically efficient means to reduce such emissions and so to mitigate climate change. A carbon tax imposes a price, the tax rate specified by the government, on each tonne of GHGs emitted, measured as tonnes of carbon dioxide equivalent (tCO₂e), but does not limit overall emissions.⁴ An emissions trading system (ETS), also called “cap-and-trade”, imposes a government-established limit on aggregate GHG emissions by specified sources, distributes tradable allowances (usually 1 tCO₂e each) approximately equal to the limit, and requires regulated emitters to submit allowances equal to their actual emissions.⁵ The market determines the price for allowances and thus the price per tCO₂e emitted.⁶

Under highly restrictive assumptions a carbon tax and an ETS yield identical results for equivalent emission reductions.⁷ These assumptions include perfect foresight (no uncertainty relating to the future), perfect competition in all markets, no interaction with other policies, and universal coverage (all sources of GHG emissions). In practice, the assumptions needed for theoretical equivalence do not hold so the performance of a carbon tax and an ETS can differ. In a given jurisdiction various factors, including taxation authority, the emissions profile, and preferences of large emitters, can affect the choice and design of an instrument. Thus, the designs of carbon taxes and GHG ETSs vary across jurisdictions.

The performance of a carbon tax or ETS as implemented is affected by some of the jurisdiction’s complementary or overlapping policies including, vehicle fuel taxes, energy efficiency standards, and renewable energy incentives, as well as external factors, such as economic recessions and world oil prices, and other events such as the shutdown of nuclear generating stations after the great east Japan earthquake and tsunami.⁸

¹ See Pollitt, 2016 section 2.5, Goldblatt, 2010 section 3 and Partnership for Market Readiness, 2017, Box 2.

² See, for example, Baranzini, et al. 2017.

³ There are numerous greenhouse gases (GHGs). Carbon dioxide (CO₂) is the most voluminous and least potent. Emissions of other GHGs are often measured in terms of tonnes of CO₂ equivalence (tCO₂e) usually based on their radiative forcing relative to CO₂ over a 100 years.

⁴ A carbon tax is a tax per tCO₂e applied to fossil fuels and/or GHG emitting activities. In this paper, taxes on fossil fuels not related to the GHG emissions due to combustion of the fuel are not considered to be carbon taxes.

⁵ The quantity of allowances distributed may be less than the established limit because some allowances may be withheld for various purposes, such as allocations to new entrants, or because they are not purchased at allowance auctions. Allowances in excess of the cap may be distributed as transitional mechanisms, such as credit for early action, or to limit price increases when allowance prices exceed specified thresholds.

⁶ As discussed below, a variant on the design is that the government specifies an emissions limit for each participant and allocates tradable credits to emitters whose actual emissions are less than their limits. Emitters that exceed their limits must buy credits to cover the excess emissions. The term “allowance” will be used to include the credits in such systems.

⁷ Weitzman 1974 and Weisbach, 2010.

⁸ OECD, 2011; Baranzini, et al., 2017.

This paper assesses the performance of operational carbon taxes and GHG ETSs implemented before the end of 2015.⁹ They are assessed for environmental effectiveness (reduction of actual emissions), cost-effectiveness (marginal abatement cost), economic efficiency, public finance and administrative issues. We do not claim observed emission reductions are due to the instrument because, as noted above, many other factors can affect actual emissions. The data and resources needed to disentangle the contribution of the instrument from the other factors were not available for this research. Nevertheless, useful insights can be drawn about the performance of carbon taxes and GHG ETSs implemented to-date.

The performance of many of the instruments implemented has been assessed from several different perspectives and with varying degrees of rigour, including: British Columbia,¹⁰ European Union ETS,¹¹ Ireland,¹² New Zealand,¹³ Norway¹⁴, Regional Greenhouse Gas Initiative (RGGI)¹⁵, Sweden¹⁶ and Switzerland.¹⁷ In addition to those assessments of individual instruments, a few studies review the performance of multiple instruments including the seven Chinese pilot ETSs,¹⁸ ETSs for greenhouse gases and other pollutants,¹⁹ carbon taxes in several European countries²⁰ and carbon taxes and GHG ETSs in 18 jurisdictions.²¹

This paper next identifies the carbon taxes and GHG ETSs implemented before the end of 2015. Then the effect of real world considerations on instrument choice and design is discussed. Criteria to evaluate the instruments are proposed in section 4. The changes in emissions subject to the carbon taxes and GHG ETSs implemented to-date are summarized in sections 5 and 6 respectively. Our assessment of the performance of these instruments in practice appears in section 7. Conclusions are drawn in section 8.

2. Carbon Taxes and GHG ETSs Implemented To-date

The carbon taxes and GHG ETSs operational as of the end of 2015 are listed in Table 1. In addition, carbon taxes have been implemented in Alberta, Chile and Colombia (2017) and ETSs were implemented in British Columbia and Fujian (2016) and Ontario and Washington (2017). Our analysis is limited to the instruments operational at the end of 2015 because data on the performance of more recently implemented instruments are not available.

⁹ Taxes and ETSs that are no longer operational, such as Australia's Carbon Pricing Mechanism, are not included in the analysis.

¹⁰ Murray and Rivers, 2015.

¹¹ Ellerman and Joskow, 2008; Ellerman, et al., 2010; Abrell, et al., 2011; Brown et al., 2012; Martin, et al., 2012; Liang, et al., 2013; and European Commission 2012, 2015 and 2017.

¹² Farrell, 2017.

¹³ Richter and Chambers, 2014; Ministry of the Environment, 2016 and Leining and Kerr, 2016.

¹⁴ Bruvoll and Larsen, 2004.

¹⁵ Hibbard, et al., 2011 and 2015.

¹⁶ Andersson, 2017.

¹⁷ FOEN, 2016.

¹⁸ Duan, et al., 2014; Dong et al., 2016 and Zhang, et al., 2017.

¹⁹ Schmalensee and Stavins, 2017.

²⁰ Andersen, 2010; and Lin and Li, 2011.

²¹ Narassimhan, et al., 2017.

Table 1. Carbon Taxes and GHG Emissions Trading Systems in Operation at the end of 2015

Carbon Taxes					GHG Emissions Trading Systems				
Jurisdiction	Year Launched	Price August 2017 US\$/tCO ₂	Share of Jurisdiction's Emissions Covered (%)	Emissions Covered by the Instrument MtCO ₂ e	Jurisdiction	Year Launched	Price August 2017 US\$/tCO ₂	Share of Jurisdiction's Emissions Covered (%)	Emissions Covered by the Instrument MtCO ₂ e
Finland	1990	69 - 73	36	21	European Union	2005	6	45	1,939
Poland	1990	<1	4	15	Alberta	2007	24	45	123
Norway	1991	4 - 56	60	32	New Zealand	2008	13	51	41
Sweden	1991	140	42	23	Switzerland	2008	7	11	17
Denmark	1992	27	45	22	RGGI (USA)	2009	4	21	86
Slovenia	1996	20	24	2	Tokyo	2010	14	20	10
Estonia	2000	2	3	4	Saitama	2011	14	18	7
Latvia	2004	5	15	1	California	2013	15	85	375
British Columbia	2008	24	70	43	Quebec	2013	15	85	68
Liechtenstein	2008	87	26	0	Beijing	2013	8	45	6
Switzerland	2008	87	35	5	Guangdong	2013	2	60	388
Ireland	2010	24	33	20	Shanghai	2013	5	57	170
Iceland	2010	12	55	3	Shenzhen	2013	6	40	30
Japan	2012	3	70	926	Tianjin	2013	1	55	100
United Kingdom	2013	24	25	127	Chongqing	2014	<1	40	126
France	2014	36	40	185	Hubei	2014	2	35	324
Mexico	2014	1 - 3	46	332	South Korea	2015	18	68	470
Portugal	2015	8	26	18					
Average		\$12.09	45.54%	Total	Average		\$7.86	48.10%	Total
Range		<1 to 140	3% to 70%	1,778	Range		<1 to 24	18% to 85%	4,280

Notes: August 2017 prices are from World Bank, Ecofys and Vivid Economics, 2017. Shares of emissions covered are from IC4E, 2017. For the EU and national jurisdictions the emissions covered by the instrument are calculated from national emissions excluding LULUCF for 2014 reported by CAIT/World Resources Institute and the share of emissions covered by the instrument. For sub-national GHG ETs the emissions covered are from the ICAP fact sheets for the respective systems. For the BC carbon tax it is calculated from the BC emissions for 2015 excluding LULUCF reported by Environment and Climate Change Canada and the share of emissions covered by the tax.

Sources: World Bank, Ecofys and Vivid Economics, 2017; Métivier et al., 2017; UNFCCC Greenhouse Gas Inventory Data, 2015; ICAP, 2017 and Environment and Climate Change Canada, 2017.

As of the end of 2015 GHG ETSs were operational in 55 jurisdictions – 34 national and 21 sub-national – while 17 national and 1 sub-national jurisdiction collected a carbon tax.²² Thus ETSs are more common even though the first carbon taxes date from 1990 while the first GHG ETS was not implemented until 2005. Of the 6,058 MtCO₂e annual emissions covered by a carbon price, ETSs cover 4,280 MtCO₂e (71%) while carbon taxes cover 1,778 MtCO₂e (29%).²³

The average share of a jurisdiction's emissions covered by an ETS (48.1%) is a little higher than the average for a carbon tax (45.5%).²⁴ The share of emissions covered ranges from 18% to 85% for ETSs and from 3% to 70% for carbon taxes. The average tax rate (US\$ 12.10) is over 50% higher than the average ETS allowance price (US\$ 7.86). The tax rates range from less than US\$ 1 to US\$ 140 while allowance prices range from less than US\$ 1 to US\$ 24.

Most of the jurisdictions with a carbon tax also have an ETS, usually covering different emissions. Thirteen of the countries with a carbon tax participate in the EU ETS. Switzerland and Alberta also have both a tax and an ETS. British Columbia has established a trading system to complement its carbon tax, but currently there are no emitters subject to the trading system. Japan and Mexico have only a tax.²⁵ New Zealand and South Korea as well as all of the sub-national jurisdictions except Alberta and British Columbia just have an ETS.

3. Instrument Choice and Design Reflect Real World Considerations

The instrument selected by a jurisdiction and its design are affected by a variety of real world considerations. Constitutional authority or political considerations may limit the choices available to a jurisdiction. Members of the European Union and European Economic Area, for example, must join the EU ETS.

In California, two-thirds of the legislature must approve a tax while legislation to implement an ETS requires only 50% approval, making an ETS easier to implement. Opponents challenged the ETS on the grounds that the auction revenue was a tax. This was rejected by the court. Legislation to extend the ETS to 2030 was passed with a greater than two-thirds majority.

An ETS needs more administrative infrastructure than a tax, especially an allowance registry and allowance market supervision, and it needs a competitive market for allowances -- a large number of participants with no dominant firms. Thus a carbon tax may be a more feasible option for a small jurisdiction or a developing country with limited administrative capacity. In some cases a small jurisdiction can address these issues by joining an ETS covering multiple jurisdictions; examples include Malta in the EU ETS and New Hampshire in the Regional Greenhouse Gas Initiative (RGGI).

²² The EU ETS covers 31 national jurisdictions and RGGI covers 9 American states.

²³ MtCO₂e = million tCO₂e.

²⁴ These are weighted averages; the share for each jurisdiction is weighted by the emissions covered by the instrument in that jurisdiction. The average tax rates and allowance prices are calculated in the same way.

²⁵ Mexico will start an ETS simulation in preparation for its pilot ETS launch in 2018. World Bank, Ecofys and Vivid Economics, 2017.

In practice a carbon tax is usually implemented as a series of taxes on various fossil fuels and/or emitting activities.²⁶ The tax rate for each fuel and activity is set so that the tax per tCO₂e emitted is the same. Tax rates of US\$ 0.0259 per litre of diesel and US\$ 0.0202 per m³ of natural gas, for example, both correspond to a tax of US\$ 10/tCO₂.²⁷ Most carbon taxes have multiple fossil fuel/activity components. Even if a fossil fuel is already taxed, a carbon tax on fossil fuel emissions may raise a number of implementation issues including coverage, tax base, reporting requirements and use of the revenue.²⁸ Taxpayers typically report the quantity of each fuel/activity subject to the tax and the tax payable, but not the associated emissions.

Existing ETSs, with the exception of New Zealand, use one of two basic designs. The more common design, the EU ETS for example, imposes a government-established cap on aggregate GHG emissions by specified sources, distributes tradable allowances (1 tCO₂e each) approximately equal to the cap, and requires regulated emitters to submit allowances or other compliance units equal to their actual emissions.²⁹ In the other design, the Tokyo ETS for example, the regulator sets an emissions limit for each participant, in effect an allocation of free allowances, a participant whose actual emissions are below its limit receives tradable credits for the difference while a participant whose emissions exceed its limit must submit purchased credits or other compliance units equal to their excess emissions.³⁰ The term “allowances” is used to include such credits.³¹ Other features, including coverage, length of the compliance period and price stability mechanisms vary widely across both designs.

Many ETSs allow participants to use credits issued for emission reductions outside the coverage of the trading program – offsets – for compliance.³² An ETS that allows the use of offsets typically specifies the activities eligible to generate such credits and usually limits the quantity that can be used for compliance purposes. Some ETSs restrict the use of offset credits to those generated in the same jurisdiction, while other ETSs allow the use of imported units. The Chinese pilots accept only offset credits generated by projects in China, while the EU ETS allows only credits issued by the Clean Development Mechanism or Joint Implementation mechanism subject to various qualitative and quantitative restrictions that have changed over time.

Once a jurisdiction selects an instrument, many design issues need to be resolved, especially the sources/emissions covered and the treatment of emissions-intensive, trade-exposed (EITE) firms that face competition from firms in jurisdictions with weaker (or no) policies to limit greenhouse gas emissions. All instruments exclude some emissions for administrative reasons – emissions that are difficult to measure, such as forests and agriculture,³³ and small emissions by numerous sources, such as

²⁶ See Partnership for Market Readiness, 2017.

²⁷ When a fossil fuel is burned virtually all of the carbon is emitted as CO₂, so it is possible to express the tax on CO₂ emissions as an equivalent tax on the carbon content of the fuel.

²⁸ For example emissions-intensive, trade-exposed (EITE) installations may be tax exempt or be eligible for refunds and the carbon tax is likely to be based on physical units while other energy taxes may be levied on other bases such as market value.

²⁹ As noted in footnote 5, the quantity of allowances distributed may be less than or greater than the cap. European Union, 2015 provides a detailed description of the EU ETS.

³⁰ This is called a “baseline and credit” system. In effect the sum of the limits of all participants is the cap on aggregate emissions. Changes to the participant population automatically adjust the cap on aggregate emissions. In contrast, in a “cap and trade” system the cap usually is not adjusted for changes to the number of sources.

³¹ This also avoids confusion with offset credits.

³² Haites, 2015.

³³ The New Zealand ETS includes forests and requires emissions reporting by agriculture.

emissions by households.³⁴ In practice, the share of a jurisdiction's emissions covered by ranges from 18% to 85% for ETSs and from 3% to 70% for taxes (see Table 1).

Imposing a carbon tax or ETS compliance obligation on an EITE firm can have adverse economic and environmental consequences. The instrument imposes a cost on the firm and makes it less competitive relative to similar firms in jurisdictions with weaker (or no) greenhouse gas emission limitation policies. Production and associated emissions may shift to the jurisdiction with the weaker policy ("leakage"), reducing the anticipated environmental benefits and imposing economic costs on the jurisdiction with the more stringent policy.³⁵ Thus virtually all instruments have provisions to protect EITE firms. In the case of taxes they include exemptions, differential tax rates, rebates and other support.³⁶ ETSs provide free allowances and rebates.³⁷

In an ETS an emitter's compliance cost is uncertain because the allowance price is not known. In practice, almost all ETSs include some price stability provisions. Typically these include the ability to use offset credits from outside the cap and/or jurisdiction for compliance and the sale of additional allowances if the price reaches a specified maximum. Some ETSs also set a minimum price. A minimum price and a maximum price together establish a "price collar".

Implementing a tax or ETS raises issues on how to treat industries whose prices are regulated, such as taxis and electric utilities. In the case of transportation fuels, the cost of the tax or ETS increases the price of the fuel and the higher fuel price typically leads to a fare increase. Electric utilities can reduce emissions by adjusting the operations of generating units that use different fuels. That makes it difficult to calculate the actual cost of a tax or ETS to a utility. During the early phases of the EU ETS provision of free allowances to electric utilities led to higher electricity prices and large profits for utilities.³⁸ To limit electricity price increases, California issues free allowances to utilities but requires that they be sold at the quarterly auctions. The revenue received is incorporated into the calculation of approved electricity prices by the regulator.

In practice, then, the instrument designs are more complex than the theoretical model with the result that it is sometimes difficult to decide whether an instrument is a tax or an ETS. For example, RGGI has a minimum price and at times between 2009 and 2013 the auction price was equal to the minimum price with some allowances not sold. During that time the ETS was in effect a tax with the tax rate equal to the minimum price. Switzerland implemented a carbon tax but allowed trading to achieve compliance. The trading option was widely used, so the tax effectively served as the non-compliance penalty for the trading system. Consistent with other studies, this paper treats both as ETSs; RGGI because the floor price prevailed only for a limited period and Switzerland because the overwhelming compliance option was trading.³⁹

³⁴ Some emissions can be included indirectly; regulating the emissions of electricity generators and residential fuel suppliers, for example, covers virtually all household emissions. Partial coverage of excluded sources is also possible by allowing them to earn offset credits for verified emission reductions.

³⁵ See Branger and Quirion, 2014 for a review. Most *ex ante* modeling studies estimate leakage rates of 5–20% (if no option to mitigate leakage is implemented), while *ex post* econometric studies have not revealed statistically significant evidence of leakage.

³⁶ Partnership for Market Readiness, 2017, Table 5, p. 18.

³⁷ European Union, 2015 outlines the provisions of the EU ETS to address leakage and lists several studies that attempt to estimate leakage for vulnerable sectors covered by the ETS.

³⁸ de Bruyn, et al., 2016.

³⁹ World Bank, Ecofys and Vivid Economics, 2017; ICAP, 2017 and Métivier, et al., 2017 also treat them as ETSs.

4. Evaluation Criteria

The main aim of this paper is to assess the performance of carbon taxes and GHG ETSs as currently implemented. That requires evaluation criteria. Drawing on criteria suggested by Goldblatt as the basis for choosing between a tax and ETS for South Africa, the following evaluation criteria are proposed.

Principal criteria:

- Environmental effectiveness – reduction of actual emissions.
- Cost-effectiveness – low marginal abatement cost.

Secondary criteria:

- Economic efficiency – low price low volatility; price signal commitment into the future to support investment decisions; and harmonization of marginal costs across jurisdictions.
- Public finance – revenue raised, cross-jurisdiction revenue flows.
- Administrative issues – institutional capacity, administrative costs, difficulty of making adjustments to rules, and minimization of corruption.

A jurisdiction implementing a carbon tax or GHG ETS may be concerned about, *inter alia*, employment, income distribution, regional impacts, economic growth and technology development. These are important issues that can be addressed through the design of the instrument and use of the revenue generated. In our judgment they are not issues inherent to a tax or ETS so we do not use them to assess the performance of the instruments.

The assessment focuses on the period from 2005 to 2016. All of the ETSs and half of the carbon taxes have been implemented during this period. Although half of the carbon taxes were implemented prior to 2005, using the same period for both taxes and ETSs means their performance is assessed in the context of the same global developments such as international climate change agreements, world fossil fuel prices and global economic conditions.

4.1 Environmental effectiveness

Since the overriding purpose of a carbon tax or GHG ETS is to reduce GHG emissions, the primary criterion is environmental effectiveness.⁴⁰ Environmental effectiveness has two dimensions – the share of the jurisdiction's GHG emissions covered by the instrument and the net reduction in aggregate emissions covered by the instrument.

All else equal, an instrument that covers a larger share of a jurisdiction's emissions should lead to larger emission reductions and so be more effective. The share of the jurisdiction's emissions covered varies widely for existing carbon taxes and ETSs as shown in Table 1 but the overall average is only slightly higher (48.1%) for ETSs than taxes (45.5%).⁴¹ Thus, any difference in environmental effectiveness depends primarily on the reduction in the aggregate emissions covered by the instrument.

⁴⁰ Some jurisdictions may have implemented an instrument, more likely a tax than an ETS, as a means of raising revenue.

⁴¹ An ETS that allows the use of offset credits for compliance extends the price signal to sources outside the cap eligible to generate credits.

Measuring emission reductions achieved by a carbon tax or ETS and comparing reductions across jurisdictions raises several issues. A tax or ETS has both price effects and revenue effects. The prices of all GHG-emitting activities and products covered by the instrument and all goods and services that use them as inputs rise motivating shifts in consumption and mitigation measures that cause emissions to fall. The revenue collected reduces income, which lowers emissions, but the revenue is then spent or distributed in ways that raise or lower emissions. These emissions impacts occur in the jurisdiction implementing the tax or ETS and in other jurisdictions.

Since these instruments put a price on GHG emissions, the price effects are the key to their performance. The first issue is whether to include the revenue effects as well. Our choice is to exclude the revenue effects for conceptual and practical reasons. The conceptual reason is that decisions on the use the revenue are distinct decisions and change over time. It is acknowledged that the proposed use of the revenue may be important in building support for a tax or ETS when it is introduced. But the use of revenue varies widely across jurisdictions and changes over time.

The practical reason is that there are several revenue effects that are difficult to estimate and that have different impacts on emissions. Revenue collected reduces spending on GHG-emitting goods and services thus lowering emissions. Some of the revenue redistributed to firms and households is spent on GHG-emitting goods and services thus raising emissions.⁴² Revenue used to fund mitigation measures lowers emissions, but measures, such as energy efficiency, that reduce emissions covered by the instrument would be double counted, as funded reductions and as lower actual emissions. The fact that these changes in emissions are spread across multiple jurisdictions increases the practical difficulties of trying to estimate the revenue effects.

Focusing on the emission reductions due to the price effects raises the issue of the appropriate geographic scope. Our choice is to focus on the emission reductions in the jurisdiction implementing the instrument. Again there are conceptual and practical reasons. The conceptual reason is that the instrument is implemented by a jurisdiction for the purpose of reducing its emissions so that is the appropriate measure of performance. The climate change impacts depend on changes to global emissions which may be lower than the domestic reductions due to leakage.

The practical reason for focusing on domestic emission reductions is that leakage is very difficult to estimate and is likely to be small in practice. Implementation of a carbon pricing instrument can lead to higher emissions by other sources – leakage – via several different channels.⁴³ Leakage is very difficult to estimate.⁴⁴ Ex ante model estimates range from 5% to 20% but lack instrument specific provisions designed to minimize leakage.⁴⁵ Ex post econometric studies, only available for Europe, find no evidence

⁴² Carl and Fedor, 2016 estimate the revenue collected in 2013 at US\$ 21.7 billion for taxes and US\$ 6.57 billion for ETSs and report the shares redistributed as tax cuts and rebates and used to fund adaptation and emission reduction measures. The shares reported for the use of tax revenue do not sum to 100.

⁴³ Zhang and Zhang, 2016 identify competitiveness effects, shifts in fossil fuel use, terms of trade effects, abatement resource effects, technology development, scale effects, and intertemporal impacts. The sources may be located in the jurisdiction implementing the policy or in other jurisdictions.

⁴⁴ Arnold, 2013; Branger and Quirion, 2014.

⁴⁵ Arnold, 2013; Branger and Quirion, 2014.

of leakage.⁴⁶ Instrument designs try to limit the leakage by reducing the economic cost to EITE sources through exemptions, rebates, free allowances or other measures.⁴⁷

The challenge, then, is to measure the changes in domestic emissions due to the price effects of the tax or ETS. The instrument reduces emissions from what they would have been. Estimating what emissions would have been and then calculating the emissions reduction achieved by the instrument is complex and speculative.

We adopt a more practical, but more stringent, measure of performance; namely the change in actual emissions subject to the instrument. Data on actual emissions are verified for compliance purposes so they are reliable and publicly available for many of the ETs. A reduction from what emissions would have been may not be sufficient to lower actual emissions so requiring a reduction in actual emissions is a more stringent measure of performance. But, it is an appropriate measure given that most of the jurisdictions that have implemented these instruments have GHG targets expressed as reductions from an historic level; 8% reduction from 1990 levels by 2020 for example.⁴⁸

The use of offset credits for compliance leads to higher actual emissions and hence poorer performance for some ETs. This applies only to ETs since none of the taxes being assessed allows the use of offsets for compliance.⁴⁹ Actual emissions are not adjusted for the use of offset credits because they were used mainly between 2011 and 2014 with negligible use during earlier and later years.⁵⁰ This means they have negligible impact on the actual emissions during the years used to calculate the rates of change in emissions.⁵¹

In summary, the environmental performance of an instrument is measured as the change in actual emissions by sources subject to the instrument. Emissions changes due to the collection and use of the revenue are excluded. Emissions changes due to the use of offsets and leakage are likewise excluded. This is a more stringent measure of performance than a reduction from what emissions would have been, but it is consistent with the targets of the jurisdictions that implemented the instruments and relies upon on much higher quality data.

⁴⁶ Branger and Quirion, 2014.

⁴⁷ Caron, et al., 2015; Lantzi, et al., 2013; Zhang, 2012; and European Union, 2015. In practice the provisions may be more generous than needed to offset the cost of the instrument. Clarke and Waschik, 2012 and Martin, et al., 2014.

⁴⁸ For example, British Columbia has a target of a 33% reduction from 2007 emissions by 2020. Quebec has a target of a 20% reduction from 1990 GHG levels by 2020. The EU target is a 14% reduction from 2005 emissions by 2020 with the ETS achieving a 21% reduction and non-ETS sectors a 10% reduction. Tokyo has a target of a 25% reduction from 2000 GHG levels by 2020. RGGI states have a regional target of a more than 50% reduction of CO₂ emissions from electricity generation from 2005 levels by 2020. The reduction target for the tax or ETS may differ from the overall target for the jurisdiction, but since these instruments cover about half of the jurisdiction's total emissions on average, an overall target that requires a reduction in actual emissions likely requires a reduction in emissions subject to the instrument as well.

⁴⁹ The Mexican carbon tax legislation allows entities to use CERs generated by CDM projects in Mexico to reduce their overall tax bill by an amount equivalent to the market value of the CERs at the time the tax is paid. It appears that these rules and valuation criteria have yet to be fully developed.

⁵⁰ Haites, 2015.

⁵¹ An assessment of the performance of the instruments on a global basis would need to address the additionality of the offset credits. See Spalding-Fecher, et al. 2012; Erickson, et al., 2014.

The performance of instruments that cover different quantities of emissions and that have been in effect for different periods of time can be compared by expressing the emission reductions as compound annual rates of reduction. The use of data on actual emissions means that the base for calculating the annual rate of reduction is the actual emissions for the first year or first compliance period the instrument was in effect and consequently, that any reductions achieved during the first year are excluded.⁵² Again this yields a conservative estimate of performance.

4.2 Cost effectiveness

Cost effectiveness focuses on achieving the emission reduction target at minimum cost. Minimizing the cost of reducing GHG emissions benefits society by freeing up resources for other priorities. Indeed the main motivation for implementing carbon taxes and GHG ETSs is that they can reduce emissions at lower cost than other forms of regulation.

The marginal abatement cost for an emitter is the tax rate (\$/tCO₂e) in the case of a carbon tax and the price of allowances in the case of an ETS. The tax rate is set by the government while the allowance price is determined by the market. Jurisdictions set the tax rate to achieve an emissions reduction target, meet a revenue target, reflect the social cost of CO₂ emissions, or match compliance costs in other jurisdictions.⁵³ The allowance price reflects the emissions target and other provisions of the ETS design.

In practice both tax rates and allowance prices span wide ranges. As shown in Table 1, tax rates range from less than US\$1 to US\$ 140/tCO₂ while allowance prices range from less than US\$ 1 to US\$ 24/tCO₂. The average tax rate (US\$ 12.09) is over 50% higher than the average ETS allowance price (US\$ 7.86).

Some analyses compare tax rates and allowance prices to estimates of the economic value of a 1 tCO₂e emission reduction (the social cost of carbon) over time.⁵⁴ This cost-benefit test is somewhat ambiguous due to the wide range of the estimated values of the social cost of carbon.⁵⁵ In addition, Morgan et al. argue that this use of the social cost of carbon is theoretically inappropriate because it incorporates numerous impacts of GHG emissions that can/should not be estimated and aggregated and applies a marginal value to a non-marginal situation; complete elimination of anthropogenic emissions.⁵⁶

Since most of the jurisdictions that have implemented a carbon tax or GHG ETS have a target that requires a reduction of actual emissions, the cost-effectiveness goal of achieving that target at the lowest cost is more appropriate than the cost-benefit test that compares the tax rate or allowance price with the social cost of carbon. Even if a specific value of the social cost of carbon could be agreed, the

⁵² The Japanese carbon tax and Irish solid fuel carbon tax took effect mid-year. The actual emissions for the first full year are used as the base of the calculation of rate of emissions reduction.

⁵³ Partnership for Market Readiness, 2017, p. 16

⁵⁴ High-Level Commission on Carbon Prices, 2017; Goulder, et al., 2017.

⁵⁵ Estimating the social cost of carbon is very complex and the results are very sensitive to the discount rate used. Stern, 2006; Smith and Braathen, 2015; US, 2016; Nordhaus, 2017; National Academies, 2017 and High Level Commission on Carbon Prices, 2017 discuss methodological issues and values used by different countries. For most carbon taxes and ETSs, the tax/allowance price per tCO₂e is lower than low estimates of the social cost of carbon (World Bank, Ecofys and Vivid Economics, 2017, Figure 7).

⁵⁶ Morgan, et al, 2017.

emission reduction that would be achieved with a tax rate or allowance price equal to that value is not known for most jurisdictions and is unlikely to coincide with its established target.

4.3 Economic efficiency

Economic efficiency has several dimensions in addition to cost-effectiveness, including price stability, a credible price signal, and harmonization across jurisdictions. The vast majority of mitigation actions involve capital investments; price stability and a credible price signal facilitate investment decisions. Harmonization across jurisdictions improves global economic efficiency. An ETS is presumed to be more countercyclical – during a recession output declines, emissions fall, allowance prices drop and compliance costs are reduced with the opposite effects during an economic boon – than a carbon tax with constant or rising tax rates, but we have found no research to document that effect.

Price stability (low volatility) facilitates evaluation of potential mitigation investments.⁵⁷ A tax is more stable than allowance prices, but changes to the tax rate tend to be unpredictable.⁵⁸ Many ETSs include provisions to enhance price stability such as sale of additional allowances when the price is high, access to external offset credits and a minimum price. Larger ETSs tend to have options and/or futures contracts that enable sources, at a cost, to obtain allowances at an agreed price at a future date thus eliminating the price uncertainty. Comparable options/contracts do not exist for taxes.

A commitment that signals future prices also facilitates mitigation investment decisions.⁵⁹ The price signal should cover a period during which most investments are repaid; at least 5 years into the future. When a new carbon tax is implemented, annual or biannual tax rate increases are often announced three to five years into the future. Most carbon taxes in effect for ten years or more have had periods of constant nominal tax rates lasting four to eight years. ETSs tend to have longer (5 to 8 years) and stronger (regulations/legislation) commitments to future caps and price stability provisions.

Global economic efficiency is improved if all sources in all jurisdictions face the same marginal abatement cost. In principle, carbon taxes can be harmonized across jurisdictions, but international coordination of a tax is very complex and has never been attempted.⁶⁰ Linking of ETSs, allowing participants to use allowances from any of the linked ETS for compliance, narrows or eliminates allowance price differences. Linking is common, but current ETSs limit the quantity of imported allowances that can be used for compliance, so price differences are narrowed rather than eliminated.⁶¹

⁵⁷ If future allowance/credit prices are uncertain, the value of future emission reductions is uncertain. Greater price volatility means greater uncertainty. Greater uncertainty favours investments with shorter payback periods.

⁵⁸ Possible mechanisms to reduce the political uncertainty associated with taxes are discussed in Partnership for Market Readiness, 2017.

⁵⁹ In their analysis of the effectiveness of feed-in-tariffs for solar photovoltaics in Europe, Dijkgraaf, et al., 2017 find the maximum effect can be as much as seven times larger with a high tariff and a long contract duration in combination with a consistent policy.

⁶⁰ Pollitt, 2016. It is the *effective* tax that must be harmonized. In addition to the carbon tax, the effective tax also depends on other taxes such as fuel taxes and VAT/sales taxes on energy, so the burden of all of those taxes on each category of emitters needs to be harmonized. Then there are practical issues of harmonizing taxes denominated in different currencies and coordinating implementation of tax changes across jurisdictions.

⁶¹ Haites, 2015.

4.4 Public finance

Both taxes and ETSs (auctioned allowances) raise revenue. The revenue can be used for a mix of economic stimulus (reduce existing distortionary taxes), equity (fair distribution of the costs of reducing GHG emissions), environmental goals (funding GHG reductions or adaptation measures) and other purposes. Although we exclude the emissions impacts of revenue collection and use, the amount of revenue collected and its use for economic and environmental purposes can be an important consideration in building and sustaining support for a carbon tax or GHG ETS.

A carbon tax imposes the tax on all designated emissions, while many ETSs distribute allowances, often most allowances, free. Combined with the fact that tax rates tend to be higher than allowance prices, carbon taxes collect more revenue than ETSs. Carl and Fedor estimate that in 2013 carbon taxes collected US\$ 21.7 billion in revenue while ETSs collected US\$ 6.57 billion.⁶²

In practice, the revenue collected from carbon taxes and ETSs is used differently. In 2013, 70% of the ETS revenue was used to subsidize “green” spending on energy efficiency, renewable energy and adaptation measures; 21% went to general revenue and 9% was used to fund corporate and/or individual tax cuts or direct rebates.⁶³ In the case of carbon taxes, 15% of the revenue was used to support green spending, 28% went to general revenue and 44% was used for tax cuts and rebates.⁶⁴ These percentages vary widely across jurisdictions for both taxes and ETSs.

A considerable body of literature is devoted to the potential economic benefits of revenue recycling -- using carbon tax or ETS auction revenue to reduce existing distortionary taxes.⁶⁵ Based on the 2013 data, as much as 85% of the tax revenue and up to 30% of the ETS revenue might be used to reduce existing distortionary taxes. These figures combine the percentages of revenue that went to general revenue (and so could offset revenue lost due to cuts in other taxes) and to tax cuts and direct rebates.

Many of the national jurisdictions that have implemented a carbon tax and/or a GHG ETS were parties to the Kyoto Protocol for the 2008-2012 commitment period, including the members of the EU ETS, Japan, New Zealand and Switzerland. The Kyoto Protocol established three international market mechanisms – the clean development mechanism, joint implementation and international emissions trading. The units traded using these mechanisms – CERs, ERUs, and AAUs respectively – were accepted, with some qualitative and quantitative restrictions, for compliance by the national ETSs of these countries.⁶⁶ CER purchases by ETS participants helped fund emission reduction measures in developing countries.⁶⁷

⁶² Although the years differ, the carbon tax revenue is almost identical to the US\$ 21.68 billion implied by the average tax rate of US\$ 12.09 and 1,778 MtCO₂ of total emissions subject to taxes shown in Table 1. This comparison is not completely fair because Carl and Fedor, 2016, include the now cancelled Australian carbon pricing mechanism in their calculation. The average allowance price of US\$ 7.86 and total emissions covered by ETSs of 4,280 MtCO₂ imply a total value of US\$ 33.64 billion. The US\$ 6.57 billion collected is about 20% of this total suggesting that overall roughly 80% of allowances are distributed free.

⁶³ Carl and Fedor, 2016.

⁶⁴ Carl and Fedor, 2016. The percentages for the use of carbon tax revenue do not sum to 100. If the shares are recalculated to sum to 100 they are 17% green spending, 32% general revenue and 51% tax cuts and rebates.

⁶⁵ See Hepburn, 2006 for a short review.

⁶⁶ Haites, 2015.

⁶⁷ The clean development mechanism, the largest of the three trading mechanisms, involved over 7,500 projects in more than 95 developing countries.

All international climate agreements have and continue to call for resource transfers from developed to developing countries.⁶⁸ With an ETS such transfers can occur via trade in allowances/offset credits as well as through government budgets.

4.5 Administrative issues

An ETS requires a more complex and more costly administrative structure than a carbon tax. Although most jurisdictions already collect taxes on fossil fuels, adding a carbon tax may involve changes to the MRV system and compliance enforcement. The administrative complexity and cost rises if the scope of the carbon tax extends beyond fossil fuels already taxed. An ETS, on the other hand, creates new administrative requirements including: an MRV system; institutions to distribute allowances, enforce compliance and monitor the allowance market; and a computer registry to track allowance holdings. Jurisdictions that have implemented an ETS, such as the EU, have offered support relating to instrument design and development of institutional capacity to other jurisdictions.

Implementation of a carbon tax or ETS usually requires specific legislation and regulations. Once implemented, changes to the tax rate or coverage can often be made as part of a budget. Changes to the rules governing an ETS tend to require changes to regulations and/or legislation which often involves a process of notice and consultation. In the case of multi-jurisdiction ETSS, such as the EU ETS and RGGI, changes can take several years to implement. No attempt has yet been made to implement a carbon tax that is harmonized across multiple jurisdictions.

We are not aware of data on the cost of implementing and administering an ETS or a carbon tax. Data on tax avoidance for carbon taxes and various possible forms of corruption for ETSS also are not available.

5. Reductions in Emissions Subject to Carbon Taxes

This section calculates changes in emissions subject to various carbon taxes and then investigates the effect of complementary policies and the contribution of carbon taxes.

5.1 Changes in emissions subject to carbon taxes

We did not find emissions data for any of the carbon taxes. That probably reflects the fact that carbon taxes usually are structured as taxes on fossil fuels and emitting activities. Governments are interested in the quantities of taxable fossil fuels and activities and the taxes due, so taxpayers are not required to report the associated CO₂ emissions.

Fortunately the emissions covered by a carbon tax can be calculated relatively easily. Since the tax rates for different fuels and activities are set to yield the same tax per tCO₂, the total carbon tax revenue collected can be divided by the tax rate per tCO₂ to get the emissions subject to the tax (see Annex 1). As

⁶⁸ Such provisions are part of the United Nations Framework Convention on Climate Change, the Kyoto Protocol and the Paris Agreement.

discussed in section 4.1, these data are actual emissions with no adjustments for leakage or emission changes due to use of the tax revenue. The compound annual rates of change use the actual emissions during the first year as the base thus excluding reductions during the first year the tax was in effect.

We were able to collect data on the carbon tax revenue collected and the carbon tax rate for 11 jurisdictions and then calculate the emissions subject to the tax from 2005 or year of implementation through 2016. The estimates for recent years match the figures for the estimates of emissions covered by the instrument presented in Table 1 quite well. We were not able to collect the requisite data for Estonia, Finland, Latvia, Liechtenstein, Mexico, Poland, or Portugal. With the exception of Finland, the tax rate is low, the emissions covered are small, or the tax has only been in effect for about two years.

Some jurisdictions including Ireland and Norway have multiple carbon tax rates for different fuels or activities. Where revenue data relating to the different rates are available, they are treated as separate taxes. Where the revenue is consolidated, we use the tax rate that generates most of the revenue or the highest tax rate. The estimated emissions may not be accurate, but the annual rates of change should be reasonably accurate. For different jurisdictions the data relate to calendar or fiscal years. Comparisons of annual rates of change are valid across jurisdictions with different fiscal years.

We ignore small changes in the coverage of the tax. For example, British Columbia offered a one-time exemption (worth C\$ 7.6 million) from the carbon tax for greenhouse operators in 2012.⁶⁹ The 2013 budget changed this to an ongoing 80% reduction of the tax for greenhouse operators. Gasoline and diesel used for agriculture were exempted from the carbon tax by the 2014 budget.

The annual rates of change in tax rates and emissions are summarized in Table 2. We have data for 11 jurisdictions of which two have two carbon taxes resulting in 13 jurisdiction/tax combinations. Of those 13 jurisdiction/tax combinations, actual emissions subject to the carbon tax fell in seven cases – Denmark, Iceland, Japan, Norway, Sweden, Switzerland, and the United Kingdom. Annual rates of emissions reduction range from less than 1% to over 47%. In the other six cases – British Columbia, Ireland (2 taxes), France, Norway (offshore activities) and Slovenia – actual emissions subject to the carbon tax increased at annual rates between 3% and 14%.

An increase in actual emissions after implementation of the carbon tax does not mean the tax had no impact; only that the tax was not sufficient to reduce actual emissions. British Columbia illustrates the point. Several estimates suggest that the tax reduced emissions by 5–15%.⁷⁰ Those estimates are reductions from what emissions would have been and some are on a per capita basis. But the BC tax has not met our more stringent criterion of a reduction in the actual emissions subject to the tax. As noted earlier, we believe our criterion is more appropriate given the province's target of reducing total GHG emissions.

⁶⁹ At the tax rate of C\$ 30/tCO₂, the exemption represented about 0.25 MtCO₂, about 0.67% of the emissions covered by the tax. Information on the magnitude of the agricultural fuel exemption is not available, but the changes in the annual emissions suggest that it is small as well.

⁷⁰ See Murray and Rivers, 2015.

Table 2. Annual Rates of Carbon Tax Increases and Change in Emissions Subject to the Tax for Selected Periods

The first three columns of data cover the period from 2005 or the inception of the tax until 2016.

Other periods are periods of declining, (virtually) constant, or rising tax rates.

Jurisdiction	Tax	Period	Average Annual Tax Rate Increase %/year	Average Annual Emission Change %/year	Period	Average Annual Tax Rate Increase %/year	Average Annual Emission Change %/year	Period	Average Annual Tax Rate Increase %/year	Average Annual Emission Change %/year
Norway	CO2-tax on mineral oil	2005-16	5.35	-0.62						
	Tax on CO2 emissions in petroleum activities on the continental shelf	2005-16	0.56	3.16	2005-07	-1.76	9.62	2008-12	0.44	-0.15
					2013-16	1.04	4.17			
Sweden	Tax on CO2	2005-16	1.99	-2.89	2005-09	3.64	-3.00	2009-14	0.00	-2.20
Denmark	Duty on CO2	2005-16	7.03	-10.28	2005-08	0.00	-0.62	2009-16	3.43	-8.92
Slovenia	CO2 tax	2005-16	2.98	6.88	2005-12	-0.02	2.63	2013-16	6.31	1.20
British Columbia	Carbon Tax	2008-16	14.72	3.62	2008-12	31.61	5.10	2012-16	0.00	2.16
Switzerland	CO2 levy on heating and process fuels	2008-16	27.54	-7.45						
Ireland	Natural gas carbon tax	2010-16	6.59	5.91						
	Solid fuel carbon tax	2014-16	4.88	13.53						
Iceland	Carbon tax	2010-15	16.11	-4.11						
Japan	CO2 tax	2013-16	44.90	-3.29						
United Kingdom	Carbon Price Floor	2014-16	91.31	-47.65						
France	Carbon tax	2014-16	77.28	5.45						

5.2 Effect of complementary policies

Almost all of the countries with a carbon tax have been part of the EU ETS since 2008.⁷¹ Since the carbon taxes are applied almost exclusively to non-ETS emissions, this creates a natural experiment; changes in non-ETS emissions can be compared for countries with and without a tax. Fourteen countries that participate in the EU ETS have a carbon tax, including countries for which we do not have data on emissions subject to the carbon tax. Three of those countries – Iceland, Liechtenstein and Norway – are not part of the EU. Sixteen countries that participate in the EU ETS do not have a carbon tax.⁷² Several of the carbon taxes were introduced after 2008; when comparing rates of emissions reduction since 2008 this favours carbon taxes.

The EU effort sharing allocations provide a rough indication of the comparability of the tax and non-tax groups in terms of their expected reductions in non-ETS emissions. The effort sharing allocations are national targets for reduction of non-ETS emissions from 2013 through 2020.⁷³ Although they are politically negotiated, the allocations reflect judgments on equity, emission reduction potential and other considerations. In aggregate the annual reductions in non-ETS emissions required from the non-tax countries exceed those of the group with taxes by -0.98% to -0.90%.⁷⁴ In short, the two groups are reasonably comparable although the non-tax countries are expected to achieve slightly larger annual reductions in their non-ETS emissions.

The EU member states are expected to meet their effort sharing allocations with EU-wide, national and sub-national policies. EU-wide policies that help reduce non-ETS emissions include CO₂ standards for road vehicles and standards for buildings and energy-related products. Member states and their sub-national jurisdictions and public and private entities are expected to implement additional measures, possibly including a carbon tax, to meet their allocation. The rates of change of non-ETS emissions for 2008-2015 were -2.08% per year for countries with a carbon tax and -2.59% per year for countries without a tax.

Non-ETS emissions then have decreased more rapidly in European countries without a carbon tax than in those with a carbon tax due, at least in part, to EU wide policies. That suggests such policies may have contributed to the reductions in emissions subject to carbon taxes shown in Table 2 for European countries. To examine that possibility we compare the annual rates of emission reductions for non-ETS emissions and for emissions subject to the carbon tax for specific countries/taxes in Table 3. The solid fuel tax in Ireland and the carbon taxes in France and the United Kingdom are excluded because they have only been implemented recently.

⁷¹ See Table 1. Japan and Switzerland have a tax but are not part of the EU ETS. Croatia is not included in the non-tax group because it did not join the EU ETS until 2013.

⁷² The non-tax countries account for roughly 57% of the non-ETS emissions of the 30 countries.

⁷³ The policy came into effect in 2013 so earlier data are not available. The 2013-2020 period is assumed to provide a better comparison of the comparability of the tax and non-tax groups.

⁷⁴ Among the countries with a carbon tax, the national allocations range from -3.08% per year for Ireland to +1.09% per year for Latvia. For non-tax countries, the national allocations range from -5.53% per year for Cyprus to +2.13% per year for Romania.

Table 3. Annual rates of change in non-ETS emissions and taxed emissions for carbon taxes for which taxed emissions could be estimated.

Country/tax	Period	Change in Non-ETS emissions	Change in taxed emissions	Share of non-ETS emissions 2015
Denmark	2008-15	-2.57%	-12.83%	64%
Norway -- Mineral oil carbon tax	2008-15	-3.40%	-2.79%	62%
Norway -- Offshore petroleum tax	2008-15	-3.40%	+1.57%	40%
Slovenia	2008-15	-2.46%	+16.94%	67%
Sweden	2008-15	-3.29%	-2.17%	63%
Switzerland	2008-15	-2.34%	-3.99%	32%
Iceland	2010-15	-10.08%	-4.11%	102%
Ireland –natural gas tax	2010-15	-0.62%	+6.58%	42%
Note Irish solid fuel tax and taxes in France and UK excluded because the history is too short.				

In only two cases, Denmark and Switzerland, have the emissions subject to the carbon tax declined more rapidly than non-ETS emissions overall. The Swiss result may be distorted by the change in scope of the tax in 2013. So the results for these individual taxes confirm the suggestion that complementary policies have probably contributed to observed emission reductions in countries that have implemented carbon taxes.

5.3 Carbon tax rates and fossil fuel prices

Why have carbon taxes had relatively little impact on actual emissions? Carbon taxes are applied mainly to transportation fuels and fossil fuels used for residential, commercial and institutional heating. The carbon tax share of diesel prices for 2005-14 for various countries is shown in Table 4. The French and Portuguese carbon taxes are not included because the diesel fuel price data series ends with 2014.

Table 4. Carbon tax as a share of diesel fuel price 2005-2014 for selected countries.

Country	Carbon tax (US\$/liter)	Diesel price (US\$/liter)	Tax as % of price
Denmark	\$0.04 to \$0.08	\$1.45 to \$1.89	2.80% to 4.42%
Finland	\$0.20 to \$0.25	\$1.26 to \$1.95	10.48% to 13.78%
Iceland	\$0.02 to \$0.03	\$1.31 to \$2.06	0.70% to 1.38%
Ireland	\$0.05 to \$0.07	\$1.35 to \$1.98	3.15% to 3.86%
Norway	\$0.08 to \$0.14	\$1.63 to \$2.11	4.41% to 6.64%
Slovenia	\$0.04 to \$0.05	\$1.21 to \$1.77	2.43% to 3.83%
Sweden	\$0.33 to \$0.43	\$1.44 to \$2.16	19.22% to 26.62%
Switzerland	\$0.03 to \$0.18	\$1.36 to \$2.06	1.93% to 9.80%

With the exceptions of Finland, Sweden and, in recent years, Switzerland the carbon tax is less than 5% of the retail price and is relatively constant over time. In addition, the carbon tax is substantially smaller than fluctuations in the retail price. In short, the carbon tax is unlikely to have a significant impact on emissions because it is a small, relatively fixed share of the price that is lost in the “noise” of price

changes. These results are consistent with the conclusions of other studies, mostly for periods ending prior to 2005, which find the emissions reductions achieved by the carbon taxes implemented by European countries are small due to the low tax rates, the modest changes in tax rates and inelastic demands for fossil fuels.⁷⁵

5.4 Changes to carbon tax rates and associated changes in emissions

Jurisdictions that implement a carbon tax must adjust the tax rate over time to achieve their emission reduction and/or other objectives. Some jurisdictions increase the tax rate on an annual or biannual basis while others have periods where the tax rate is stable. To examine the relationship between changes in tax rates and changes in emissions we define periods with declining (1), stable (4) and rising (14) tax rates. The annual rates of tax rate and emissions changes for each of these periods by jurisdiction and tax are shown in Table 2. The periods range in length from 3 to 12 years.

The emissions changes are calculated for the same periods as the tax rate changes. Since most carbon taxes apply to fossil fuels an increase in the tax rate manifests itself as a higher price for consumers. The price elasticity of demand for fossil fuel uses is highly inelastic in the short run and elastic in the long run.⁷⁶ This means that it takes some time for the emission reductions due to a tax rate (price) increase to be fully realized. Hence, the time period for which emission reductions are calculated should lag the time period during which rates rise, but we do not know the appropriate lag. The longer the time period, the larger share of the emissions reductions that occur during the same period.⁷⁷

Higher rates of tax increases would be expected to lead to greater emission reductions. The annual rates of changes to tax rates and emissions listed in Table 2 are shown in Figure 1. The expected relationship would be a curve sloping down to the right. For declining or stable tax rates (near the origin) emissions might increase (be above the horizontal axis). Indeed, emissions often rise with declining, stable and small increases in tax rates, but emission reductions have also occurred. For high rates of tax increase, annual emissions would be expected to decline (be below the horizontal axis). Again, both emissions increases and reductions have occurred during periods of substantial tax rate increases.

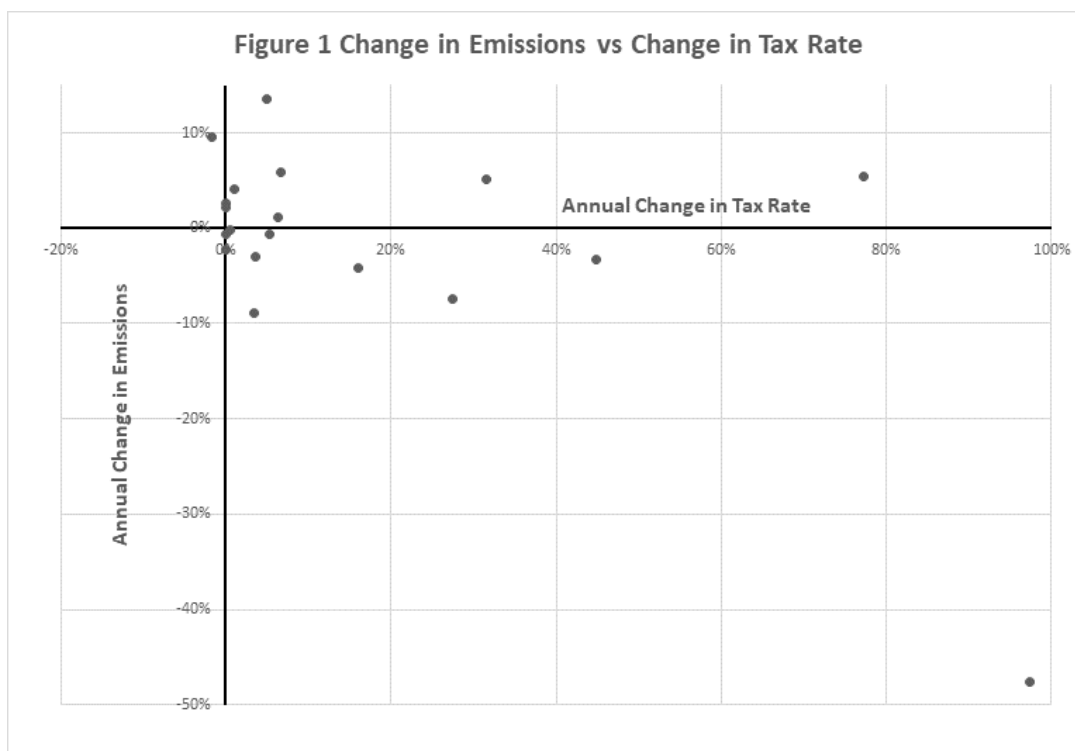
Clearly, the data relating to the existing carbon taxes do not fall neatly into the expected pattern. Imperfect association of emission changes with tax rate changes, as noted, likely is part of the explanation. Factors, such as complementary or overlapping policies, economic recessions, world oil prices and other events can have a significant impact on a jurisdiction's emissions and so distort the expected pattern. Finally, the expected pattern assumes that consumers respond promptly to energy price increases. But the impact of the tax rate changes on energy prices may be small. And research into the adoption of energy efficiency measures indicates that many market barriers inhibit adjustments to energy price increases.⁷⁸

⁷⁵ See Bruvold and Larsen, 2004; Andersen, 2010; Lin and Li, 2011; and Andersson, 2017.

⁷⁶ Phoumin and Kimura, 2014; Labandeira, et al., 2016

⁷⁷ If the lag is 2 years, a 3 year period only captures the full effect of the tax rate change during the first year, but a 12 year period captures the full impact for the tax rate changes during the first 10 years.

⁷⁸ Gillingham and Palmer, 2014; Gerarden, et al, 2017; Ameli and Brandt, 2015.



Until research yields better insight into the relationship between tax rate changes and emissions changes, administrators will need to adjust tax rates on an *ad hoc* basis balancing politically acceptable tax rate increases with emission reduction goals.

6. Reduction in Emissions Subject to GHG ETSs

This section calculates changes in emissions covered by various GHG ETSs and then considers the relationship between changes in emissions caps and changes in emissions.

6.1 Changes in emissions covered by GHG ETSs

Most ETSs report aggregate data on the allowances and offsets surrendered and financial payments made for compliance. We used that information to calculate the verified emissions covered by the ETSs.⁷⁹ As discussed in section 4.1, the amounts are actual emissions with no adjustments for leakage, use of offset credits or emission changes due to use of the ETS revenue. In a few jurisdictions – California, European Union, Quebec and Switzerland – there have been significant changes in the scope of the ETS. These are treated as separate ETSs for purposes of the analysis. The aviation component of the EU ETS likewise is treated as a separate ETS because it has its own allowances.

⁷⁹ Except for the New Zealand ETS, the verified emissions, in tCO₂e, are equal to the sum of the units surrendered plus the financial payments. In the New Zealand ETS one unit was required for emissions of two tCO₂e, so the sum was doubled to get the verified emissions.

The annual rates of reduction are calculated using the actual emissions for the first year as the base because pre-ETS emissions data are not always available and where available are sometimes considered to be inflated. This approach excludes any reductions during the first year the ETS was in effect. Annual emissions data sometimes are not available for ETSs with multi-year compliance periods such as RGGI; then we use the data for the first period as the base for calculating the second period reductions.

The annual rates of change in emissions covered by the ETSs are summarized in Table 5. The data cover 45 jurisdictions; the seven Chinese pilots are not included because they treat emissions data as confidential, Quebec only reports data by compliance period and in South Korea the first compliance period is still underway. We exclude California from the table because the rates of change are calculated over one or two years and hence may not be indicative. The data for the two regional ETSs – EU ETS and RGGI – are aggregate, rather than jurisdiction-specific, emissions. So we have data for eight ETSs, but for two of them there two periods with different coverage yielding a total of ten jurisdiction/ETS combinations.

Table 5. Annual Rates of Change in Emissions Caps and Emissions Covered by GHG ETSs for Selected Periods

Jurisdiction	Period	Average Annual Change in Emissions Cap %/year	Average Annual Emission Change %/year	Period	Average Annual Change in Emissions Cap %/year	Average Annual Emission Change %/year
European Union	2008-12	0	-1.61	2012-16	-1.87	-4.14
EU ETS Aviation	2012-16	0	-7.58			
Alberta	2007-13	NA	2.51			
New Zealand	2011-16	NA	6.50			
Switzerland	2008-12	1.48	-1.98	2013-16	-0.72	-4.52
RGGI	2012-14	-9.27	-6.40			
Tokyo*	2010-15	NA	-3.03			
Saitama*	2011-14	NA	-1.36			

Notes: * These are rates of change for energy use because the carbon content of electricity is assumed to be constant. The carbon content of electricity rose as a result of greater reliance on fossil-fired generation due to the shutdown of most nuclear generators after the 2011 great east Japan earthquake and tsunami.

NA = not applicable

Sources: Calculated from data from individual ETSs for allowances/offsets surrendered, emissions caps, and banked allowances.

Actual emissions have fallen in at least six of the ETSs covering 41 jurisdictions with annual rates of emission reductions up to 6.4%. Emissions increased in Alberta and New Zealand and possibly in Tokyo and Saitama. The fact that actual emissions increased does not mean the ETS had no impact on emissions, just that the impact was not sufficient to reduce actual emissions given other developments.

The Alberta Specified Gas Emitters Regulation (SGER) applies to installations with emissions in excess of 100 ktCO₂e. An installation subject to the regulation has an emissions limit equal to its production during the year and an emissions intensity 12% below its baseline (20% as of 2017). The growth in total emissions is due in part to an increase in the number of installations subject to the regulation.⁸⁰

The New Zealand ETS does not have an emissions cap. Participants are required to submit eligible domestic and international allowances and offsets equal to their actual emissions. Until 2015 there was no limit on the use of international allowances and offsets. Beginning in 2010, the prices of those units fell significantly with the result that they accounted for over 95% of the units surrendered for compliance from 2012 through 2014.⁸¹ Use of international allowances and offsets was banned after 2014.⁸²

The Tokyo and Saitama ETSs claim emission reductions, but actual emissions may have increased. Electricity accounts for a large share of the emissions covered by these ETSs and they assume a constant carbon intensity for electricity.⁸³ The great east Japan earthquake and tsunami of 2011 led to the gradual shutdown of all nuclear generating stations and greater reliance on fossil-fired generation. As a result, the carbon intensity of electricity for Tokyo and Saitama rose. Unfortunately, the share of total emissions due to electricity use is not available for either ETS so the impact on actual emissions cannot be calculated. Entity-specific emissions data for 985 facilities covered by the Tokyo ETS from 2010 through 2015 suggest a small increase in actual emissions due to the change in the generation mix.⁸⁴

California is excluded from Table 5 because the time periods over which emission changes can be calculated are very short so the trends may not be meaningful. Although it has multi-year compliance periods, California publishes annual emissions of entities subject to its ETS. Our standard approach of using the first year as the base for calculating emission changes yields only a one year change (a 0.40% increase) for the first compliance period and (a 1.68% decline) for the second period. Using data for the year prior to the start of each phase yields two year annual reductions of -4.81% and -2.79% respectively for the two compliance periods.

The ETS designs vary widely in terms of compliance period, price stability mechanisms, use of offsets, and other features yet actual emissions have fallen in most cases. Complementary or overlapping policies including, vehicle fuel taxes, energy efficiency standards, and renewable energy incentives, likely contributed to the emission reductions by entities participating in the ETSs.⁸⁵ Annual data for the EU and Swiss ETSs, for example, show that the 2008-2009 financial crisis and recession led to a drop in emissions during 2009 followed by a recovery in 2010 before continuing to decline.

⁸⁰ The Alberta SGER will be replaced by an output-based emissions allocation system in 2018.

⁸¹ Haites, 2015.

⁸² The ETS is being restructured to set the supply of allowances on a rolling five year basis consistent with New Zealand's emissions target, introduce auctioning, and phase-in a one unit per tCO₂ compliance obligation between 2017 and 2019. See <http://www.mfe.govt.nz/nzets/about-nz-ets-review>

⁸³ The reported reductions are changes in energy use, rather than emissions.

⁸⁴ An estimated increase of 0.35% per year.

⁸⁵ Data covering 28 members of the EU ETS for the period 2008 through 2016 indicate annual rates of emissions change between +1.78% and -7.84% with an overall average of -2.89%. All of these countries had the same ETS design and allowance prices during that period, so the different rates reflect differences in the mix of participants, local economic conditions and complementary policies.

6.2 Changes in emissions and emissions caps

Jurisdictions that implement a GHG ETS must limit the emissions by participants. When an ETS is first implemented the emissions limit tends to be generous for several reasons:

- Participants lobby for a limit that requires minimal reductions to keep compliance costs low;
- Equitable treatment of participants that implemented emission reductions prior to the start of the ETS often takes the form of early action credits or allocations that exceed actual emissions; and
- The government may not have accurate data on historic emissions.

Consequently actual emissions are often less than the quantity of allowances distributed leading to accumulation of a bank of unused allowances. The size of the allowance bank at any given time is difficult to determine – information on the units distributed is usually available but the quantity needed for compliance is not known for the same date.⁸⁶ Estimates of the size of the allowance bank for different ETSs are provided in Table 6. All of the ETSs have accumulated allowance banks. The banks range in size from 15% to over 500% (5 years) of the annual compliance obligation.

Table 6. Estimates of Allowance Banks Accumulated by ETSs

Jurisdiction	Period	Bank*	Notes
EU ETS	2005-07	>15%	Surplus allowances not valid for post 2007 compliance
	2008-16	>100%	Use of Kyoto Protocol units restricted
Switzerland	2008-12	>100%	Severe restrictions on post 2012 use of surplus 2008-12 units
	2013-16	>50%	Use of Kyoto Protocol units restricted
RGGI	2009-11	>15%	Unallocated and unsold allowances withdrawn
	2012-14	>50%	Emissions cap reduced by over 25%
New Zealand	2010-16	>500%	Use of Kyoto Protocol units banned after 2014. Bank peaked at over 600% in 2015
California and Quebec	2013-14	>35%	
Tokyo	2010-15	>15%	
Notes: * Bank at the end of the period is expressed as a percentage of the annual compliance obligation – annual emissions except for New Zealand where the compliance obligation is one unit for two tCO ₂ emitted. 100% means that the banked units are equal to 1 year's compliance obligation.			

ETSs have implemented several measures to reduce the number of banked allowances, including:

- Effective cancellation of surplus allowances – cancellation of surplus European Union allowances (EUAs) at the end of Phase 1 of the EU ETS, severe restrictions on the quantity of 2008-2012 Swiss allowances (CHUs) that can be used for 2013-2020 compliance by the Swiss ETS and withdrawal of unallocated and unsold units by state governments participating in RGGI;
- Severe restrictions on compliance use of external offset credits – after 2012 compliance use of Kyoto Protocol units was severely restricted by the EU and Swiss ETSs and New Zealand prohibited their use after 2014.

⁸⁶ The EU ETS, for example, distributes most allowances for 2017 before installations complete 2016 compliance, so the number of surplus units after 2016 compliance is difficult to calculate.

- Larger reductions to the emissions cap – a reduction in the emissions cap by RGGI after 2009-2012 and a shift from stable emissions limits to annual reductions in the limit by the California, EU, Quebec, RGGI and Swiss ETSS.
- Delayed distribution of allowances – the EU ETS “backloading” provision deferred the distribution of allowances from 2014-2016 to 2019-2020.

In addition to the measures to reduce the allowance banks mentioned in Table 6, ETS designs have changed in ways that should reduce the accumulation of allowance banks. Many ETSS now have declining annual emissions caps, although the caps are declining more slowly than emissions except for the RGGI system (see Table 5). Market stability reserves to manage the size of allowance banks have been adopted by California-Quebec, the EU and RGGI systems. While their designs differ they all withdraw allowances from the market when the supply is abundant and release allowances if shortages arise. Data on their effectiveness are not yet available.

Governments have clearly been cautious in setting the limits for their ETSS. However, emission reductions have occurred in most of the ETSS despite the accumulation of surplus allowances. In effect participants assume that the ETS will continue and that the allowances will be eligible for future compliance use despite instances where surplus allowances have been discounted or cancelled.⁸⁷

7. Comparison of Actual Carbon Taxes and GHG ETSS

Existing carbon taxes and GHG ETSS are compared in Table 7 using the criteria proposed in section 4.

ETSS perform better than carbon taxes on the principal criteria. Actual emission reductions are more common for ETSS and the marginal cost is generally substantially lower. On average ETSS cover a slightly larger share of the jurisdiction’s GHG emissions.

That assessment is subject to several caveats:

- Some carbon taxes achieve emission reductions at marginal costs comparable to those of ETSS. And some ETSS have not achieved emission reductions.
- The observed emission reductions can not be attributed solely to the ETS or tax. Evidence presented in section 5.2 suggests that complementary policies contributed significantly to the reductions of non-ETS emissions achieved by European countries with carbon taxes. Complementary policies and other developments likewise may have helped reduce emission covered by GHG ETSS.⁸⁸
- Several ETSS and taxes are not included in the comparison due to lack of data.

Comparison of changes in non-ETS emissions in European countries with and without a carbon tax suggest that the carbon taxes have had a limited impact in most of those jurisdictions. Carbon tax exemptions, the relatively low tax rates and modest changes to the tax rate limit the reduction of actual emissions achieved by the carbon tax. Most of those carbon taxes apply to residential, commercial and transport use of fossil fuels, uses where consumers may not respond promptly or fully to the price changes induced by the tax.

⁸⁷ Cancellation or discounting distributed allowances is an economic loss for the allowance owners, but the financial loss may be small since most allowances are distributed free.

⁸⁸ Bang et al., 2017.

Table 7. Comparison of existing carbon taxes and GHG ETSs

Criterion	Carbon Taxes	GHG ETSs
Principal Criteria		
Environmental effectiveness: Reduction of actual emissions in the jurisdiction	7 of 13 jurisdiction/tax combinations have reduced actual emissions since 2005 or the start of the tax but complementary policies are probably responsible for the reductions in 3 of those jurisdictions	At least 6 of 10 ETSs covering 41 jurisdictions have reduced actual emissions since inception of the ETS
	Share of the jurisdiction's emissions covered ranges from 3% to 70% with a weighted average of 45.54%	Share of the jurisdiction's emissions covered ranges from 11% to 85% with a weighted average of 48.10%
Cost-effectiveness: low marginal abatement cost	Tax rates range from <US\$1 to US\$140 with a weighted average of US\$12.09 per tCO ₂	Allowance/credit prices range from <US\$1 to US\$24 with a weighted average of US\$7.86 per tCO ₂
Secondary criteria		
Economic efficiency: Low price volatility	Tax rates are stable but can change with little notice	Many ETSs include provisions to enhance price stability and larger ETSs tend to have options and/or futures contracts that enable sources to obtain allowances at an agreed price at a future date
Economic efficiency: price signal commitment into the future	Commitments to future tax rates tend to be limited to 3 to 5 years often when the tax is introduced	Commitments to cap reductions and price stability provisions often span 5 to 8 years and with emission reduction targets to 2030 or 2050
Economic efficiency: harmonization of marginal costs across jurisdictions	Harmonization of tax rates is very complex and has never been attempted	Linking ETSs is common but almost always subject to quantitative limits so price convergence is limited
Public finance: revenue raised	Carbon taxes raise far more revenue than ETSs. Most tax revenue is used for tax cuts and rebates or general revenue. Uses vary widely by jurisdiction and change over time.	ETSs raise less revenue than taxes because in many ETSs most allowances are distributed free to EITE sources. Most revenue is used for green purposes, but this varies widely.
Public finance: cross-jurisdiction revenue flows	Cross-jurisdiction revenue flows occur through government budgets	In addition to government transfers linking ETSs can lead to cross-jurisdiction revenue flows
Administrative issues: institutional capacity	Carbon taxes are usually relatively easy to implement because most jurisdictions already collect taxes on fossil fuels	An ETS creates several new institutional requirements
Administrative issues: administrative costs	Administrative costs are usually lower for carbon taxes	Administrative costs are usually higher for ETSs
Administrative issues: difficulty of making adjustments to rules	Changes to the tax rate or coverage can typically be made as part of a budget	Changes to the rules governing an ETS tend to require changes to regulations and/or legislation which often involves a process of notice and consultation
Administrative issues: minimization of corruption	Data on tax avoidance are not available	Information on the scale of different forms of corruption for ETSs is not available.

More research is needed to improve instrument design. In the case of a carbon tax, it is conceptually clear that the tax rate must rise over time to adjust for inflation, increased income, technological change and other factors to achieve further emission reductions. The huge variation in tax rate changes and rates of emission reduction/increase indicate that this relationship is not well understood in practice.

- Denmark, Iceland, Ireland, Norway and Switzerland regularly increase the tax rate at one to three year intervals.
- British Columbia, France, Japan, and the United Kingdom have introduced carbon taxes with rapid (over 30%/year) tax rate increases for periods of three to five years. Such rapid tax increases probably are not sustainable. In British Columbia they were followed by a five year rate freeze. What will happen in the other jurisdictions is not yet known.

Many of the ETSs accumulate significant quantities of unused (banked) allowances and offset credits. A growing allowance bank indicates that emissions can be reduced faster than the rate at which the cap is declining. Since a growing bank depresses the market price, additional reductions can be achieved at relatively low cost. Some ETSs already have, and others are implementing, a market stability reserve to remove excess allowances from the market when prices are low and release additional allowances into the market when prices are high. The impacts of those of those mechanisms will need to be assessed.

Complementary policies are alleged to be a complete waste of resources in the case of an ETS because the cap does not change so any reductions achieved can be fully offset by higher emissions within the ETS.⁸⁹ That is conceptually correct, but subject to several qualifications in practice. First, almost all ETSs have experienced over-compliance so reductions achieved by complementary policies are small or not being fully offset. Second, to the extent that complementary policies contribute to over-compliance they tend to lead to more rapid reduction of the emission cap. Third, over-compliance leads to removal/cancellation of surplus allowances which captures some of the reductions due to complementary policies. Fourth, complementary policies can address market barriers that do not respond well to price signals. Finally, complementary policies may yield reductions outside the ETS cap.

In the case of a carbon tax complementary policies can address market barriers that do not respond well to price signals and may yield reductions by sources not subject to the tax. Resources are wasted if the marginal cost of reductions achieved by the complementary policy is higher than the tax rate.

With respect to the economic efficiency criteria, carbon taxes provide better short-term price stability but poorer long-term price signals than ETSs. But price stability provisions and, where available, forward contracts and options, reduce short-term price volatility for ETSs. A carbon tax that has a track record of annual or biannual rate adjustments provides a longer term price signal comparable to that of the five to eight year cap reduction commitments of many ETSs.

Harmonization of tax rates or allowance prices only occurs where multiple jurisdictions are fully linked or part of the same ETS, as in the case of the EU ETS, RGGI and California/Quebec (and Ontario from 2018). Other ETSs have links but they are subject to qualitative and quantitative restrictions which limit price convergence. Harmonization of tax rates would be very complex in practice and has not been attempted.

Carbon taxes raise much more revenue than ETSs. Indeed that has been an important motivation for implementing a carbon tax in some jurisdictions. Many ETSs distribute most of their allowances free for

⁸⁹ Schmalensee and Stavins, 2017.

two reasons; to ease the cost of adjusting to the new regulation and to protect the competitiveness of EITE sources. Free allowances provide an incentive to reduce emissions but reduce the financial cost of compliance.⁹⁰ Carbon taxes address these issues with exemptions, differential tax rates, and rebates. At least 44% of the carbon tax revenue is used for tax cuts and rebates.⁹¹ EITE exemptions might contribute to the lower share of emissions covered by taxes.

The ability of ETSs to facilitate cross jurisdictional financial transfers via allowance/offset purchases is interesting, but ETSs that allowed such imports, including the EU, Switzerland and New Zealand, have severely restricted their use and new ETSs in China and South Korea prohibit or tightly control imports of allowances and offsets from foreign jurisdictions.

A carbon tax is usually easier to implement and less costly to administer than an ETS. An ETS requires a new administrative infrastructure including: an emissions monitoring, reporting and verification (MRV) system; institutions to distribute allowances, enforce compliance and monitor the allowance market; and a computer registry to track allowance holdings. A carbon tax is usually applied to fossil fuels which are already subject to various taxes. Adding a carbon tax may involve changes to the MRV system and compliance enforcement. Extending the carbon tax beyond fossil fuels increases its complexity and administrative cost. Although the administrative costs of an ETS are generally considered to be higher than those for a carbon tax, the proliferation of ETSs indicates administrative costs are not a significant barrier.

It is interesting to note that ETSs are much more common for sub-national jurisdictions than carbon taxes, despite the extra administrative capacity they require. The sub-national ETSs are found in Canada, China and the US and the choice of an ETS might be influenced by the constitutional distribution of taxing authority in those countries.

8. Conclusions

Carbon taxes and GHG ETSs are becoming increasingly common. The first carbon taxes date from 1990 and the first GHG ETS from 2005. Since 2005 implementation of ETSs has spread more rapidly. They now cover substantially more jurisdictions and more emissions than carbon taxes. The designs vary widely. The instrument choice and design are affected by circumstances unique to the jurisdiction, such as its emissions profile, constitutional provisions, economic structure and other policies.

At the end of 2015, 17 GHG ETSs were operational in 55 jurisdictions while 18 jurisdictions collected a carbon tax. Despite the number of taxes and ETSs implemented and the fact that some have been in effect for over a decade, little is known about their actual performance.

Our review indicates that both instruments can yield emission reductions, but emission reductions are not guaranteed. Further research is needed to identify the characteristics of an effective tax and ETS.

⁹⁰ A source whose emissions exceed its free allocation must purchase allowances to cover the excess. A source whose emissions are less than its free allocation can sell its surplus allowances.

⁹¹ Carl and Fedor, 2016. The percentages reported do not sum to 100. When the percentages are calculated to sum to 100, roughly 50% goes to tax cuts and rebates. Rebates probably go to entities, such as EITE firms, subject to the tax while tax cuts are reductions of other taxes. Carl and Fedor group even though the public finance and economic impacts are quite different.

Emission reductions are more common for ETSs than taxes and on average allowance prices are lower than carbon tax rates. The observed emission reductions can not be attributed solely to the ETS or tax. Complementary policies and other developments affect the emissions covered by the instrument. The effects of the 2008-09 economic recession and the nuclear plant shutdown after the great east Japan earthquake and tsunami are evident in the data. More rigorous analysis is needed to determine the emission reductions that can be attributed to the instruments.

Interesting patterns for which we do not have an explanation include:

- Most carbon taxes are implemented by national jurisdictions in countries that also have an ETS – countries that are part of the EU ETS and Switzerland. The fact that the EU and Swiss ETSs do not cover residential, commercial and transportation fossil fuel use may be part of the explanation.
- Despite the greater administrative infrastructure required by an ETS, they are much more common than taxes for sub-national jurisdictions. That may reflect the constitutional distribution of taxing authority in Canada, China and the US. In any case the extra administrative demands are not a significant practical barrier to implementation of an ETS.

Topics that warrant further research to improve the effectiveness of these instruments include:

- The most effective combination of a carbon tax and a GHG ETS since many jurisdictions have implemented both instruments.
- The characteristics of a tax rate strategy that leads to actual emission reductions; the level of the tax rate needed to have an impact on emissions and the frequency and magnitude of tax rate changes needed to achieve continual reductions efficiently are not known.
- The most effective treatment of EITE sources under each instrument; is allocation of free allowances under an ETS more effective than an exemption/rebate under a tax? And since many jurisdictions have both a tax and an ETS, which should cover the EITE sources?
- The design features of an effective ETS; do a commitment to a declining annual cap, price stability provisions, market stability reserve, links with other ETSs, or other features have a significant impact on the emission reductions achieved?
- The cost and contribution of complementary policies. Do they yield emission reductions beyond those achieved by the tax or ETS and at what cost?

It is clear that both instruments can reduce emissions. Given the body of evidence available it is time to learn from that experience to make both instruments more effective so that they can better contribute to the effort to reduce global GHG emissions.

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Annex 1: Carbon Tax Revenue, Carbon Tax Rate and Estimated Emissions Covered by the Tax by Jurisdiction, 2005 through 2016														
Jurisdiction	Currency	Tax	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Norway	NOK (million)	CO2-tax on mineral oil	3899	4366	4469	4453	4442	4682	4719	4349	4913	5167	5965	6464
	NOK/tCO2	Tax rate	195	199	203	207	214	218	222	225	229	330	338	346
		Emissions (MtCO2)	19.9949	21.9397	22.0148	21.5121	20.757	21.4771	21.2568	19.3289	21.4541	15.6576	17.6479	18.6821
	NOK (million)	Tax on CO2 emissions in petroleum	3417	3373	3963	2254	2133	2207	2182	2280	4387	4743	5061	5116
	NOK/tCO2	Tax rate	430	421	415	225	231	232	229	229	443	445	453	457
		Emissions (MtCO2)	7.94651	8.01188	9.5494	10.0178	9.23377	9.51293	9.52838	9.95633	9.90293	10.6584	11.1722	11.1947
		Total emissions (MtCO2)	27.9414	29.9516	31.5642	31.5299	29.9908	30.99	30.7851	29.2852	31.3571	26.316	28.8201	29.8768
Sweden	SEK (million)	Tax on CO2	25535	24745	25127	25639	26084	27334	25369	25243	24031	23334	24605	22957
	SEK/tCO2	Tax rate	910	920	930	1010	1050	1050	1050	1050	1050	1050	1130	1130
		Emissions (MtCO2)	28.0604	26.8967	27.0183	25.3851	24.8419	26.0324	24.161	24.041	22.8867	22.2229	21.7743	20.3159
Denmark	DKK (million)	Duty on CO2	5151.2	5085.7	5108.4	5056.4	5005.7	5821.9	5881.7	5709.3	5761.6	3523.67	3651.56	3296.52
	DKK/tCO2	Tax rate	90	90	90	90	150	153	156	160	164	167	170	190
		Emissions (MtCO2)	57.2356	56.5078	56.76	56.1822	33.3713	38.0516	37.7032	35.6831	35.1317	21.0998	21.4798	17.3501
Slovenia	EUR (million)	CO2 tax	45.931	33.53	27.341	30	30	31	29	55	106	106	124	132
	Euro/tCO2	Tax rate	12.52	12.52	12.5	12.5	12.5	12.5	12.5	12.5	14.4	14.4	17.28	17.3
		Emissions (MtCO2)	3.66861	2.67812	2.18728	2.4	2.4	2.48	2.32	4.4	7.36111	7.36111	7.17593	7.63006
British Columbia	CAD (million)	Carbon Tax				306	542	741	959	1120	1222	1198	1190	1220
	CAD/tCO2	Tax rate				10	15	20	25	30	30	30	30	30
		Emissions (MtCO2)				30.6	36.1333	37.05	38.36	37.3333	40.7333	39.9333	39.6667	40.6667
Switzerland	CHF (million)	CO2 levy on heating and process fuels				220	217	588.6	498	552	642	757.78	827.022	829
	CHF/tCO2	Tax rate				12	24	36	36	36	48	60	60	84
		Emissions (MtCO2)				18.3333	9.04167	16.35	13.8333	15.3333	13.375	12.6297	13.7837	9.86905

Jurisdiction	Currency	Tax	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Ireland	Euro (million)	Natural gas carbon tax						196.05	298.231	353.954	381.061	368.129	395.48	405.816
	Euro/tCO2	Tax rate						15	17.5	20	20	20	22	22
		Emissions (MtCO2)						13.07	17.0418	17.6977	19.0531	18.4065	17.9764	18.4462
	Euro (million)	Solid fuel carbon tax									7.316	17.233	23.496	24.432
	Euro/tCO2	Tax rate									10	10	11	11
		Emissions (MtCO2)									0.7316	1.7233	2.136	2.22109
		Total emissions (MtCO2)						13.07	17.0418	17.6977	19.7847	20.1298	20.1124	20.6673
Iceland	ISK (million)	Carbon tax						1914	2062	3061	3013	3004	3274	
	ISK/tCO2	Tax rate						545	817.5	1090	1090	1120	1150	1190
		Emissions (MtCO2)						3.51193	2.52232	2.80826	2.76422	2.68214	2.84696	0
Japan	JPY (million)	CO2 tax								25800.2	88598.7	162166	164493	243769
	JPY/tCO2	Tax rate								95	95	192	192	289
		Emissions (MtCO2)								271.581	932.618	844.615	856.736	843.491
United Kingdom	GBP (million)	Carbon Price Floor										1023	1208	1026
	GBP/tCO2	Tax rate Carbon Price Floor										4.94	9.55	18.08
		Emissions (MtCO2)										207.085	126.492	56.7478
France	EUR (million)	Carbon tax										325.44	2500	4000
	Euro/tCO2	Tax rate										7	14.5	22
		Emissions (MtCO2)										46.4914	172.414	181.818
Note: Data are a mix of calendar and fiscal years. Shaded areas indicate (virtually) constant tax rates for three years or more.														
Sources: Compiled from financial accounts of the respective jurisdictions, State and Trends of the Carbon Market 2014 through 2017, OECD database on Policy Instruments for the Environment available at: https://pinedatabase.oecd.org/ and other miscellaneous sources.														